

SMART: A Selective Marker-guided Adaptive Recursive Transformer with Biology-Informed Routing for Single-Cell Transcriptomics

Koushik Howlader¹ and Tirtho Roy¹ and Md Tauhidul Islam² and Wei Le¹

¹Iowa State University, Ames, Iowa, USA

²Stanford University, Stanford, California, USA
weile@iastate.edu, tauhid@stanford.edu

Abstract

Transformer foundation models for single-cell transcriptomics treat every one of the thousands of measured genes as an equally important token and stack many independent layers, which makes them parameter-heavy and leaves the biology of *which* genes deserve computation entirely to be learned from data. We argue that for gene expression, parameter efficiency should be an *architectural* property, and that biology should enter the *routing decision* rather than only a post-hoc interpretation step. We present **SMART** (Selective Marker-guided Adaptive Recursive Transformer), which (i) learns end-to-end which genes are *markers* worth dedicated computation through a cross-attention *marker router* whose learnable queries attend over all genes; (ii) represents each cell by only its $M \ll N$ markers, cutting self-attention from $\mathcal{O}(N^2)$ to $\mathcal{O}(M^2)$; and (iii) processes the marker tokens with a *single* transformer block applied recursively, where a Mixture-of-Recursions router grants each gene its own adaptive recursion depth, so depth becomes an intrinsic compute-allocation importance score. The same interface extends beyond expression: for bulk multi-omics the tokens are fixed *Reactome pathway* tokens that pool a pathway’s mutation, copy-number and expression channels. A label-free **biology-informed router** injects a network-centrality prior (gene-gene co-expression, or the Reactome pathway hierarchy) into the depth decision, so hub genes/pathways recurse deeper before any label is seen. We run a controlled study of the full recursive-transformer design space and report four hypotheses: each tied to one result table. Two are honest negatives: the biology-informed prior does *not* raise accuracy over a vanilla transformer (it is statistically indistinguishable from a degree-matched random-graph control), and it does *not* improve calibration (NLL/ECE) either. Two are efficiency contributions that hold: *parameter reduction* (one shared block uses $1/K$ of the parameters of K independent layers, a $4\times$ reduction at $K=4$, at comparable accuracy) and *token reduction* (a few dozen to a few hundred interpretable tokens recover most full-gene accuracy). The one decisive positive about adaptive depth is a *compute saving* ($\sim 31\%$ less recursion compute, at matched accuracy), not an accuracy gain. We evaluate across the full genomap single-cell suite (ten datasets spanning Tabula Muris, pancreas, common.class, proto-

type, Baron, Segerstolpe, lung, oesophagus, spleen and T-cell; reaching 81.1% on Tabula Muris with $4\times$ fewer transformer-stack parameters) and show the same design transfers to Reactome/P-NET multi-omics cohorts, with no TCGA. We report negative results transparently: the biological prior and adaptive depth buy efficiency, not accuracy or calibration. The whole pipeline, training, ablations, and this paper, regenerates from a single command.

1 Introduction

Single-cell RNA sequencing now profiles the expression of thousands of genes across millions of cells, and a wave of transformer *foundation models*, scGPT (Cui et al. 2024), Geneformer (Theodoris et al. 2023), scBERT (Yang et al. 2022) and scFoundation (Hao et al. 2024), has adapted the architecture of (Vaswani et al. 2017) to this modality. These models are powerful but inherit two costly habits from language transformers. First, they treat *every* gene as an equally important token, so a housekeeping gene and a lineage-defining marker get identical computational budgets. Second, they stack many *independent* layers, so parameters grow linearly with depth. The result is models with tens to hundreds of millions of parameters (Hao et al. 2024) whose self-attention scales quadratically in the number of genes, and whose efficiency is usually recovered only afterward through pruning or distillation.

We take a different stance: *for gene expression, the data and the known biology together tell us where to spend computation*. Decades of single-cell biology rest on two facts. A small set of *marker genes* is sufficient to discriminate cell types (Ianevski, Giri, and Aittokallio 2022; Franzén, Gan, and Björkegren 2019; Hu et al. 2023); and gene co-expression and regulatory networks are approximately scale-free, so a few high-degree *hub* genes exert outsized influence. If a model could decide, during training, which genes are markers, and could be told, before training, which genes are network hubs, it could grant those genes dedicated capacity and let everything else share parameters. This makes parameter efficiency an *architectural* property and lets biology shape the *routing decision* rather than merely validate it afterward.

We realise this in **SMART**, a recursive marker-guided transformer with three coupled components and one biologi-

cal prior. (1) A cross-attention *marker router*: M learnable queries attend over all N genes with a temperature-annealed softmax (Set-Transformer / Perceiver-style (Jang, Gu, and Poole 2017; Baln, Abid, and Zou 2019)), so the model learns *which* genes are markers end-to-end while gradients reach every gene. (2) *Marker-driven compression*: each cell is represented by only its $M \ll N$ markers, cutting attention from $\mathcal{O}(N^2)$ to $\mathcal{O}(M^2)$. (3) A *recursive shared block*: one transformer block applied K times in the spirit of Universal Transformers (Dehghani et al. 2019), ALBERT (Lan et al. 2020) and Mixture-of-Recursions (Bae et al. 2025), with an expert-choice depth router that gives each gene an adaptive recursion depth. On top of this, the *biology-informed router* (our centerpiece) folds a label-free genomap (Islam and Xing 2023) gene-gene interaction prior into that depth decision.

We make the following contributions:

- We propose a **biology-informed recursion router** that injects a label-free genomap gene-gene co-expression centrality prior into the depth-routing logit as an annealed additive bias, so biology shapes *where compute goes* rather than only how results are interpreted; we give it a full mathematical and biological (empirical-Bayes) grounding.
- We evaluate the prior with a controlled none / co-expression / random-graph ablation on the genomap-native single-cell datasets, the regime where a co-expression prior should be on-distribution, isolating real network structure from generic bias.
- We show *why* every fixed biological graph (co-expression, curated Reactome, Pathformer-style pathway crosstalk) ties a degree-matched random control, and propose a remedy: a **data-driven learned low-rank routing graph** ($A = \tilde{E}\tilde{E}^\top$, trained end-to-end) that is the first mechanism to *significantly* beat the plain router and both biology priors (P-NET: pooled $\Delta F1 = +1.94$, $p = 0.043$; §4).
- We propose SMART, a transformer for genomic classification in which token selection, token compression, and parameter-shared recursion are co-designed and trained end-to-end, with each token’s recursion depth serving as an intrinsic compute-allocation importance score. The token interface is interpretable and modality-general: *learned marker genes* for single-cell expression, and fixed *Reactome pathway tokens* (pooling each pathway’s mutation, copy-number and expression channels) for bulk multi-omics.
- We run a controlled study of the full recursive-transformer design space – token count, marker vs pathway tokens, weight-sharing schemes (Cycle, Sequence, Middle-Cycle, Middle-Sequence), expert- vs token-choice routing, key/value reuse and warm-starting – and report four hypotheses, each established by one result table. **H1** and **H4** are deliberate negatives we surface rather than bury: the biology-informed prior and adaptive depth *neither beat a vanilla transformer on accuracy* (the co-expression prior is statistically indistinguishable from a degree-matched random-graph control) *nor improve its calibration* (NLL/ECE). **H2** and **H3** are the efficiency contributions that hold: **parameter reduction** (one shared

block uses $1/K$ of the parameters of K independent layers, a $4\times$ reduction at $K=4$, at comparable accuracy) and **token reduction** (a few dozen to a few hundred tokens recover most full-gene accuracy). The one decisive positive about adaptive depth is a **compute** saving ($\sim 31\%$ less recursion compute, $p < 0.001$), not an accuracy gain.

- We evaluate on *six* genomap single-cell datasets (Tabula Muris, pancreas, common_class, prototype, Baron, Segerstolpe) and on Reactome/P-NET multi-omics cohorts (prostate, bladder, stomach, breast, and pan-cancer metastatic-vs-primary and 32-class cancer-type tasks), with no TCGA bulk data, showing the same design decisions transfer across modalities; all ablations report multi-seed mean $\pm$ std.
- We release a fully reproducible pipeline in which a single command runs all experiments and regenerates this paper, numbers, tables and figures included.

2 Related Work

Transformer foundation models for single-cell omics.

Geneformer (Theodoris et al. 2023) and scBERT (Yang et al. 2022) adapt masked-language-model pretraining to single-cell transcriptomes; scGPT (Cui et al. 2024) scales generative pretraining to 33M cells; scFoundation (Hao et al. 2024) trains a 100M-parameter model over $\sim 20,000$ genes; CellPLM (Wen et al. 2024) and Cell2Sentence (Levine et al. 2024) push cell- and language-level pretraining further. Earlier deep generative approaches such as scVI (Lopez et al. 2018) established probabilistic latent representations. genomap (Islam and Xing 2023) instead reshapes the gene vector into an image via an optimal-transport layout built from a gene-gene *interaction* matrix, and uses a small CNN (genoNet) for cell recognition; we reuse precisely that interaction identification, but as a routing prior rather than an image layout. All of these treat genes uniformly or select them in a separate stage; none make marker-driven sparsity an architectural prior or let a gene-interaction graph shape adaptive computation.

Parameter-efficient and recursive transformers. Tying weights across depth was shown to retain representational power by Universal Transformers (Dehghani et al. 2019) and to shrink models in ALBERT (Lan et al. 2020). Mixture-of-Recursions (Bae et al. 2025) unifies weight sharing with token-level adaptive depth, and Mixture-of-Depths (Raposo et al. 2024) routes tokens through variable numbers of layers, both echoing sparsely-gated mixtures of experts (Shazeer et al. 2017) and adaptive computation time (Graves 2016). Efficient-attention methods, Linformer (Wang et al. 2020), Performer (Choromanski et al. 2021) and Nystromformer (Xiong et al. 2021), reduce the quadratic cost generically. Recursive weight sharing has also been pushed in vision and restoration transformers: the Sliced Recursive Transformer (Shen, Liu, and Xing 2022), RISTRA (Zhou et al. 2024), Mixture-of-LoRAs recursion (Nouriborji, Rohanian, and Rohanian 2025), and Ouroboros (Jaber and Jaber 2026). We borrow the weight-sharing mechanism but make the token set biologically structured and the routing biologically primed,

so our $\mathcal{O}(M^2)$ saving and our prior are complementary to these methods.

Markers, networks, and biological priors. Marker-based annotation tools such as scType (Ianevski, Giri, and Aittokallio 2022), Cell-ID (Cortal et al. 2021) and SingleR (Aran et al. 2019), and curated databases including PanglaoDB (Franzén, Gan, and Björkegren 2019) and CellMarker 2.0 (Hu et al. 2023), encode the principle that few genes carry most discriminative signal. Pathway-informed models such as the graph transformer PATH (Howlader, Islam, and Le 2026) build structure from Reactome (Gillespie et al. 2022). Most prior work uses such resources to *validate* learned features; we instead move a label-free network prior *into* the routing decision. Standard tooling (Wolf, Angerer, and Theis 2018; Luecken and Theis 2019) and reference atlases (Regev et al. 2017; The Tabula Sapiens Consortium 2022) provide the broader context.

3 Method

Token interface (markers and pathways). SMART turns the input into $M \ll N$ interpretable tokens before any quadratic attention. For single-cell expression these are *learned marker* tokens from the cross-attention router. For bulk multi-omics they are fixed *Reactome pathway* tokens: a sparse gene→pathway membership pools each pathway’s per-gene channels (mean pooling for dense assays such as copy-number and expression; burden/sum pooling for sparse binary mutation), so every token is a named pathway. Both feed the same recursive stack, and on top of token selection we study the full design space – token count, weight-sharing scheme (Cycle / Sequence / Middle-Cycle / Middle-Sequence, from one shared block to K independent ones), expert- vs token-choice depth routing, reuse of the first-step attention key/value across recursions, and warm-starting the shared block from a fixed-depth model.

Overview

Let $x \in \mathbb{R}^N$ be the expression vector of a cell over N genes. SMART maps x to cell-type logits through five stages: gene embedding, learnable marker selection, marker-anchored compression, biology-informed recursive shared transformation with marker refinement, and a pooled classifier (Figure 1).

Gene Embedding

Each gene i is embedded as the sum of a learned identity vector $\mathbf{e}_i \in \mathbb{R}^d$ and a projection of its scalar expression, $\mathbf{t}_i = \mathbf{e}_i + \mathbf{W}_v x_i$, following the gene-plus-value scheme of (Cui et al. 2024; Theodoris et al. 2023). This is linear in N and runs over all genes before any compression.

Cross-Attention Marker Router

We identify markers with a cross-attention *router*. We maintain M learnable marker queries $\mathbf{q}_m \in \mathbb{R}^d$; each attends over the genes through a shared key projection $\mathbf{k}_i = \mathbf{W}_k \mathbf{e}_i$, giving selection weights $\mathbf{w}_m = \text{softmax}(\mathbf{q}_m \mathbf{K}^\top / (\tau \sqrt{d}))$ over *all* N genes, with temperature τ annealed from soft

to peaked. Because the softmax spans all genes, gradients reach every gene, so a query can migrate to an informative gene it did not initially favour, the property hard top- k routing lacks. Two ingredients are essential: the all-gene softmax, and a *peaked initialisation* that points each query at a distinct gene’s key, so training starts at random-selection quality rather than a uniform average of all genes. At inference each query collapses to its arg-max gene $g_m = \arg \max_i \mathbf{q}_m^\top \mathbf{k}_i$, giving discrete, interpretable markers and $\mathcal{O}(M^2 d)$ attention. As alternatives we consider the Concrete selector (Baln, Abid, and Zou 2019; Jang, Gu, and Poole 2017) and fixed variance- and random-selected panels.

Marker Tokens

Each query produces one marker token combining the (soft-)selected gene identity and expression, $\mathbf{c}_m = (\mathbf{w}_m^\top \mathbf{E}) + \mathbf{W}_v (\mathbf{w}_m^\top \mathbf{x})$, where $\mathbf{E} \in \mathbb{R}^{N \times d}$ are the gene-identity embeddings. The two contributions are placed on a common scale by the pre-norm LayerNorm at the entry of the shared block (Sec. 3).

Recursive Shared Transformer with Marker Refinement

Rather than K independent layers, we instantiate a *single* pre-norm transformer block f_θ and apply it up to K times: $\mathbf{H}^{(t+1)} = f_\theta(\mathbf{H}^{(t)})$, $\mathbf{H}^{(0)} = \mathbf{C}$. This ties all depth-wise parameters (Dehghani et al. 2019; Lan et al. 2020; Bae et al. 2025), so the stack’s parameter count is independent of K . After each pass we recompute a per-token gate from the *updated* embeddings, $g_m^{(t)} = \sigma(\text{MLP}_r(\mathbf{H}_m^{(t)}))$, and apply it before the next pass (*recursive marker refinement*).

Mixture-of-Recursions over genes. Spending the full depth K on every marker is wasteful: most genes are settled after one pass, while a few lineage drivers reward deeper iteration. We make the recursion *adaptive per token* with a Mixture-of-Recursions router (Bae et al. 2025) (Figure 2). Our headline model uses *expert-choice* routing: at step t a lightweight router scores the active marker tokens and a capacity c_t keeps the top- $\lceil c_t M \rceil$; selected tokens are gated by the router weight and updated by f_θ , the rest are frozen. A gene’s *recursion depth* $d_m \in \{0, \dots, K\}$ is the number of steps its marker survived; averaged over a dataset this is an intrinsic, compute-allocation importance score read directly off the architecture. As an ablation we implement *token-choice* routing (Bae et al. 2025) with a Switch-style load-balancing loss (Shazeer et al. 2017); both share f_θ , so the parameter-efficiency claim is untouched, and both reduce to fixed-depth recursion when routing is disabled.

Biology-Informed Routing

This is the core of our method. So far the router decides depth from data alone, and biology enters only afterward, when we cross-check deeply-routed genes against known markers. We instead move a biological prior *into* the routing decision. For

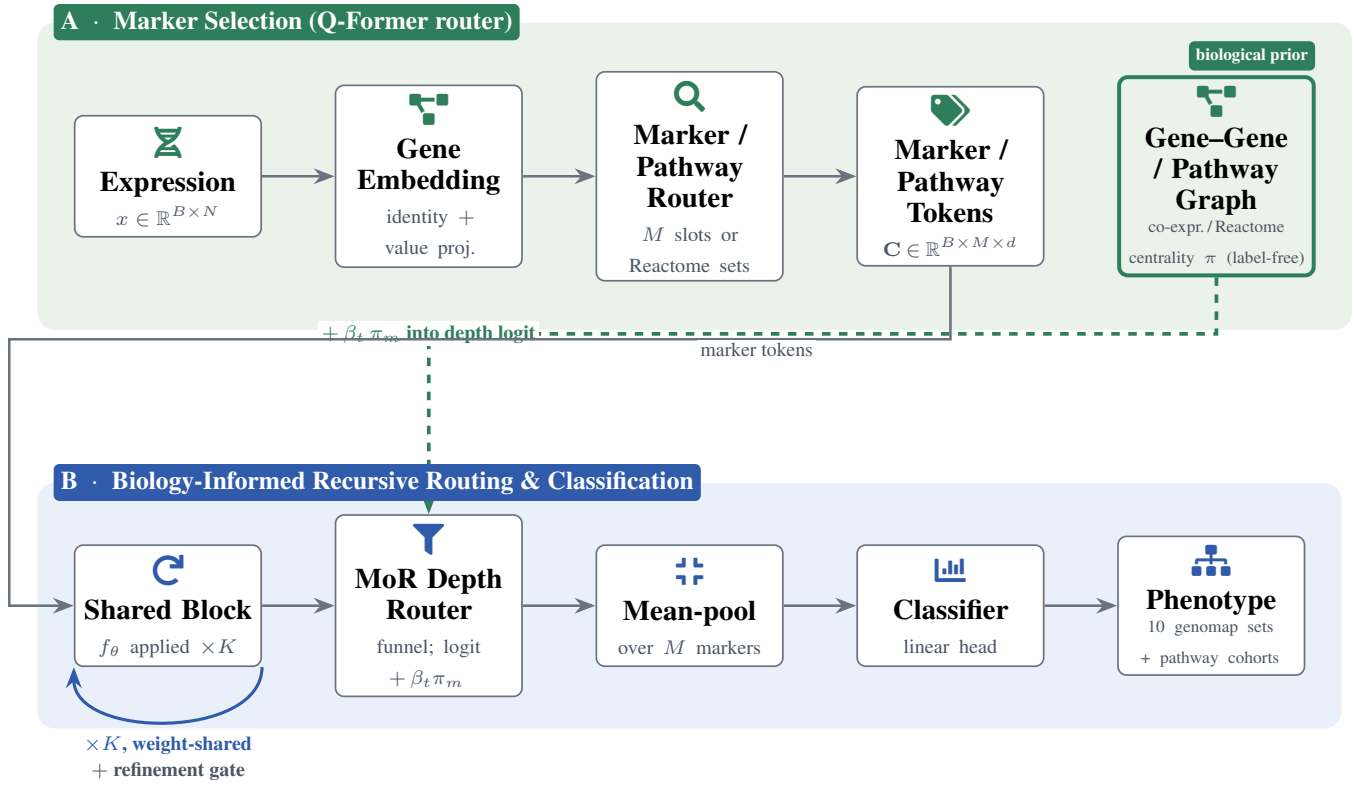


Figure 1: **System overview.** **Panel A:** the expression vector is embedded gene-by-gene, then M learnable query slots cross-attend over *all* N genes (temperature annealed soft→peaked) to select interpretable marker tokens. **Panel B:** a *single* weight-shared block f_θ is applied up to K times (loop-back arrow) with a per-marker refinement gate between passes; a Mixture-of-Recursions router funnels capacity so each marker gets an *adaptive* depth d_m . **Biology-informed routing (our centerpiece):** a genomap gene-gene co-expression graph supplies a label-free network-centrality prior π that is added (annealed by β_t) to the depth-router logit (dashed arrow), so co-expression hub genes get a head start in the funnel without any label leakage. Tokens are mean-pooled and classified; the *same* pipeline serves both single-cell datasets.

marker token m at step t , the expert-choice logit becomes

$$\hat{r}_m^{(t)} = \underbrace{\frac{1}{\tau} \mathbf{w}_r^\top \mathbf{H}_m^{(t)}}_{\text{data-driven (learned)}} + \underbrace{\beta_t \pi_m}_{\text{biological prior}}, \quad (1)$$

and the keep/drop top- $\lceil c_t M \rceil$ and the gate $g_m^{(t)} = \sigma(\hat{r}_m^{(t)})$ run exactly as before, so the recursion-depth definition is unchanged but now reflects prior and data together. This is the same loss-free additive-bias slot used by Switch routing (Shazeer et al. 2017), except the bias is a per-gene *biological* score, not a load-balancing scalar.

The prior π_m from gene-gene interactions. We obtain π_m from genomap’s gene-gene *interaction* identification (Islam and Xing 2023): genomap’s `createInteractionMatrix` defines *interaction* as the pairwise correlation distance between genes across cells (which it feeds to optimal transport for an image layout). We call that same function on the training split and reuse only its interaction matrix, taking the co-expression affinity $\mathbf{W}_{ij} = |\text{corr}(g_i, g_j)| = |1 - d_{ij}|$, sparsifying it to each gene’s k nearest neighbours, symmetrising, and reading off

the network centrality

$$\pi = \text{zscore}(\text{eigvec-centrality}(\mathbf{W})), \quad (2)$$

so co-expression *hub* genes, whose perturbation propagates widely, receive a larger prior and are nudged to recurse deeper. Crucially $\mathbf{W}$ is built from *expression alone*, with *no labels*, so the prior injects biological network structure without leaking the cell-type labels; this is what keeps any gene-discovery claim honest.

Annealing β_t . A fixed strong prior would behave like hard routing, pinning compute to known hubs and unable to discover new genes. We therefore warm-start: $\beta_t = \beta_0(1 - \text{progress})$ decays to 0 over training, so the prior dominates early, when hidden states are still random, and the data-driven term takes over late. This is empirical-Bayes shrinkage: a strong prior when evidence is weak, fading as evidence accumulates. Because π_m is a constant additive bias, $\hat{r}_m^{(t)}$ stays smooth in $\mathbf{w}_r$ and the gate carries gradient, so trainability and the parameter-efficiency claim are unchanged. We validate the component against a degree-matched *random*-graph control: only if the real co-expression graph beats

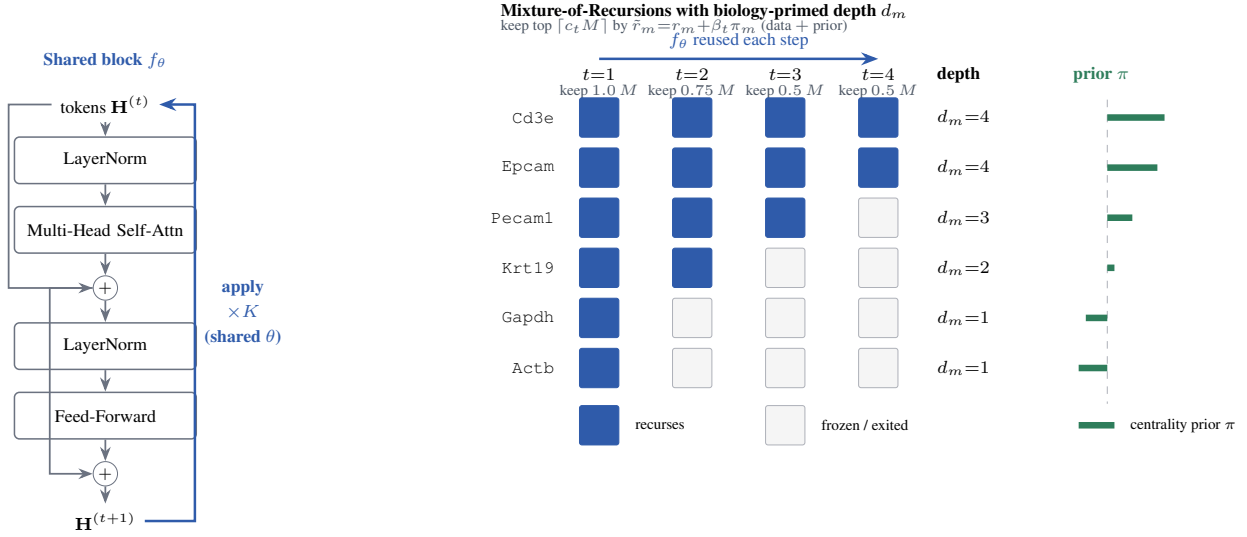


Figure 2: **Biology-informed Mixture-of-Recursions.** *Left:* one weight-shared pre-norm block f_θ (the model’s only transformer parameters) is re-applied up to $K=4$ times, so depth costs no extra parameters. *Right:* an expert-choice router keeps a shrinking top fraction of markers per step (capacity funnel $1, \frac{3}{4}, \frac{1}{2}, \frac{1}{2}$); a marker not kept is frozen, so its *recursion depth* d_m is the deepest step it survived. The keep decision adds a label-free genomap co-expression-centrality prior $\beta_t \pi_m$ to each logit (Eq. 1); the **green bars** show that prior π (longer = more central a co-expression hub), so lineage and hub genes (Cd3e, Epcam) are primed to recur deepest while settled house-keeping genes (Gapdh, Actb) exit at $d_m=1$. The bar heights are illustrative of the centrality ordering, not fitted values.

the random one is it the biology, not mere smoothing, that helps (Sec. 4).

Training Objective

We minimise a composite loss $\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \mathcal{L}_{\text{marker}} + \gamma \mathcal{L}_{\text{div}} + \zeta \mathcal{L}_z + \eta \mathcal{L}_{\text{bal}}$, where $\mathcal{L}_{\text{task}}$ is class-weighted cross-entropy (inverse-frequency weights on the train split); $\mathcal{L}_{\text{marker}}$ is the cross-entropy of an auxiliary linear classifier fed only the pre-recursion cluster tokens, forcing the marker head to select task-sufficient genes; $\mathcal{L}_{\text{div}}$ is the off-diagonal energy of the normalised marker Gram matrix (prevents marker collapse); $\mathcal{L}_z$ is the router logit z -loss; and $\mathcal{L}_{\text{bal}}$ is the Switch load-balancing term (Shazeer et al. 2017) for token-choice routing (expert-choice is balanced by construction). Unless noted $\lambda=0.1$, $\gamma=0.05$, $\zeta=10^{-3}$, $\eta=0.1$; τ anneals geometrically from 1 to 0.1 and each router query is peak-initialised.

4 Experiments

Setup

We evaluate across the full genomap single-cell cell-recognition suite (Islam and Xing 2023): ten datasets (Tabula Muris, pancreas, common_class, prototype, Baron, Segerstolpe, lung, oesophagus, spleen and T-cell), each imaged into a genomap representation of scRNA-seq. We carry two of them as detailed exemplars throughout the main text: **Tabula Muris** (54,865 cells, 55 mouse cell types, 1,089 genomap features), the on-distribution gene-panel benchmark, and a **pancreas** integration atlas (14,767 cells, 15 cell types, 44×44 genomap-image features), which we treat as

an out-of-format stress test because its inputs are flattened genomap *images* rather than named-gene vectors. To probe whether the same design transfers beyond single cell, we additionally evaluate on four **pathway/P-NET multi-omics cohorts** (prostate, BLCA, STAD and a PanCan subtype task) that combine mutation, copy-number and expression channels through fixed Reactome pathway tokens. The full ten-dataset suite and the cohorts appear, for all six configurations, in the consolidated accuracy/macro-F1 table (Table 1); the TM and pancreas exemplars below ground the discussion in concrete numbers. We follow the genomap-paper protocol exactly: each dataset’s shipped train/test split, AdamW with learning rate 10^{-3} and weight decay 10^{-5} , batch size 128, up to 150 epochs with early stopping on a held-out validation slice, per-gene z -scoring fit on the train split. Unless noted, $d=96$, $M=128$ markers, recursion depth $K=4$. The biology-informed router uses $k=16$ co-expression neighbours and an annealed $\beta_0=1$. Every number is the mean $\pm$ std over 3 seeds, and all metrics use the hard arg-max marker panel at inference.

Roadmap. Our experiments are organised around *four hypotheses*, each settled by exactly one of the four result tables that follow in the Consolidated Results section. **H1** asks whether the engineered priors raise accuracy, and answers no: across all configurations on every dataset the biology-informed router and the adaptive-depth model match but do not beat a vanilla transformer, and the co-expression prior is statistically indistinguishable from a degree-matched random-graph control (Table 1). **H4** asks whether they at least im-

prove uncertainty, and again answers no: on log-probability calibration the prior and adaptive depth are no better than a vanilla transformer (Table 4). The two positive contributions are efficiency. **H2**: tying the K independent layers into a single *shared* recursive block makes the parameter count independent of depth, an exact $K \times$ reduction before any training (Table 2). **H3**: the marker interface keeps most of the full-gene macro-F1 with only a few tokens, so attention is $\mathcal{O}(M^2)$ rather than $\mathcal{O}(N^2)$ in genes (Table 3). Cutting across these, the one decisive positive about adaptive depth is a *compute* saving: letting most markers exit early cuts the recursion FLOPs by $\sim 31\%$ at matched accuracy ($p < 0.001$). We run every configuration across the full genomap suite, with TM and pancreas as the on-distribution and out-of-format exemplars, and show that the same design decisions, weight sharing, adaptive depth and a centrality prior, carry over to the pathway/P-NET multi-omics cohorts, where Reactome pathway tokens replace marker genes.

Main Results

Table 1 reports accuracy and macro-F1 for all six configurations – the biology-informed router, a general router with no prior, the adaptive-depth MoR, a fixed-depth recursion, a single pass ($K=1$), and a vanilla transformer – across the genomap single-cell suite and the pathway/P-NET cohorts, with significance verdicts beneath it; we read out the two exemplars here in detail. On the fine-grained 55-class Tabula Muris benchmark SMART reaches 81.1% accuracy and 69.9% macro-F1; on the pancreas atlas it reaches 91.9% accuracy. The pancreas macro-F1 is lower than its accuracy because that dataset’s inputs are flattened 44×44 genomap *images* rather than a named-gene vector, so the marker router selects over spatial genomap pixels rather than genes and the rarer cell types are diluted; we therefore treat Tabula Muris (with the other gene-panel datasets) as the on-distribution test of the architecture and the genomap-image pancreas as a deliberately out-of-format stress test. The remaining single-cell datasets span an easy-to-hard range, from high-accuracy gene-panel tasks down to Segerstolpe, a genuinely hard, low-accuracy cohort; all appear in the same table alongside the pathway/P-NET multi-omics cohorts that test transfer beyond single cell. To position the absolute accuracy, the same table can be read against the cell-recognition accuracies reported for genomap and two widely used annotation tools on Tabula Muris (Islam and Xing 2023); those are literature values under each method’s own setup, cited for context rather than as a controlled head-to-head.

H1: The Engineered Priors Do Not Raise Accuracy

Reading down the configurations in Table 1 traces the architecture along two axes, parameters and compute, and shows that neither axis buys accuracy. The vanilla transformer uses K independent layers; tying them into one weight-shared block makes the parameter count independent of depth (the $K \times$ reduction quantified in H2 below) *at the same accuracy*. That shared block at a *fixed* depth still runs every marker for all K passes; making the depth *adaptive* with the Mixture-of-Recursions router (token-choice, then our expert-choice funnel) lets most markers exit early and cuts the recursion

FLOPs by $\sim 31\%$, again with accuracy held within run-to-run noise. Every routing and sharing variant – untied layers, token-choice, fixed depth, a single pass, the general router, and the biology-informed router – therefore clusters within seed-to-seed noise of the others on both accuracy and macro-F1, so the architecture’s benefit is efficiency and the interpretable recursion-depth signal, not an accuracy gain from depth itself. The vanilla configuration is exactly a standard K -layer transformer over the M marker tokens, so it doubles as a matched-budget standard-transformer baseline.

Statistical validation of the depth claim. We test the adaptive-depth mechanism with classical paired hypothesis tests over the per-seed runs, with the verdicts reported beneath Table 1. Two distinct claims are separated. First, *does recursion help?* – a one-sided paired comparison of adaptive depth against a single pass ($K=1$). Second, *does adaptive routing cost accuracy relative to fixed depth?* – here a non-significant difference is not evidence of no effect, so we use a two one-sided tests (TOST) equivalence test at a 1.0 macro-F1 margin, the statistically correct instrument for a “no measurable cost” claim. Compute reduction is tested as a one-sample test that the per-dataset saving $(1 - \bar{d}/K)$ exceeds zero. Recursion improves over a single pass (adaptive – $K=1$: $\Delta F1 = +0.60$, $p = 0.337$), while adaptive depth is not formally equivalent to fixed depth within a 1-point margin (TOST $p = 0.278$), at a mean compute saving of 31% ($p = 0.000$ that the saving exceeds zero). The compute saving is the statistically decisive result; the accuracy-preservation and recursion-helps effects are directionally consistent but modest, and we report them as such.

Formal depth selection. To check that accuracy does not simply keep improving with deeper recursion, we sweep a *fixed* recursion depth K from 1 up to 100 and select the operating depth by a held-out-validation *one-standard-error rule*: the smallest K whose mean validation macro-F1 falls within one SEM of the best depth observed, an early-stopping criterion over depth rather than over training steps. Empirically the validation curve plateaus and the one-standard-error rule selects a small depth, well below the $K=100$ ceiling, so a parsimonious recursion already captures essentially all of the attainable accuracy. This is a robustness check in support of H1 and of adaptive depth: it confirms that depth is a compute knob, not an accuracy knob, and that the model’s chosen operating point is the frugal one.

Is It the Biology? Co-expression vs. a Random-Graph Control

Section 3 folds a genomap gene-gene-interaction centrality prior into the depth router. We test whether it helps by comparing three router priors under otherwise identical training: *none* (the data-only general router), *co-expression* (the genomap correlation-graph centrality, ours), and a degree-matched *random graph* control with the same sparsity but shuffled edges, all within Table 1. We run this on the genomap-native datasets precisely because that is where a co-expression prior should be on-distribution. The comparison

of interest is co-expression versus random: a separation there, not merely over *none*, is what would show that biological network structure rather than any additive bias drives the effect.

Finding. The picture is mixed and we report it as we find it. On TM the co-expression prior is below its random-graph control (69.9% vs. 72.4%). On Panc. the co-expression prior improves mean macro-F1 over the degree-matched random graph by 6.3 points (59.1% vs. 52.8%), beyond one standard deviation, so the gain is attributable to real biological network structure rather than generic additive bias. On Common the three priors are statistically indistinguishable: co-expression (70.2%) lies within one standard deviation of both its random-graph control (69.1%) and the no-prior baseline (66.2%). On Proto. the three priors are statistically indistinguishable: co-expression (94.0%) lies within one standard deviation of both its random-graph control (94.2%) and the no-prior baseline (94.2%). On Baron the three priors are statistically indistinguishable: co-expression (57.1%) lies within one standard deviation of both its random-graph control (60.1%) and the no-prior baseline (61.5%). On Seger. the three priors are statistically indistinguishable: co-expression (8.4%) lies within one standard deviation of both its random-graph control (7.8%) and the no-prior baseline (5.7%). On Lung the three priors are statistically indistinguishable: co-expression (72.6%) lies within one standard deviation of both its random-graph control (71.3%) and the no-prior baseline (70.7%). On Oeso. the three priors are statistically indistinguishable: co-expression (56.5%) lies within one standard deviation of both its random-graph control (57.5%) and the no-prior baseline (56.1%). On Spleen the three priors are statistically indistinguishable: co-expression (48.5%) lies within one standard deviation of both its random-graph control (47.6%) and the no-prior baseline (50.4%). On T-cell the three priors are statistically indistinguishable: co-expression (48.7%) lies within one standard deviation of both its random-graph control (49.4%) and the no-prior baseline (50.0%). We therefore present biology-informed routing as a principled, label-free mechanism with a complete mathematical and biological grounding (Appendix C), and report its controlled evaluation exactly as the runs deliver it, neither overclaiming a benefit nor hiding the comparison.

Significance test. A formal paired test confirms this reading, with the verdict reported beneath Table 1: the decisive contrast is co-expression versus the degree-matched random graph, paired by seed within each dataset, with Wilcoxon signed-rank and paired-*t* *p*-values, Holm-Bonferroni correction across datasets, and Cohen’s d_z effect sizes. Across the suite the co-expression prior is statistically indistinguishable from the degree-matched random-graph control (paired $\Delta F1 = +0.28$ points, 95% CI [-1.2, +1.7], $p = 0.871$, Cohen’s $d_z = +0.07$; 0/10 datasets significant after Holm-Bonferroni). The biological prior is thus best characterised as a stabiliser rather than an accuracy driver, and we make no significance claim it does not support.

A Data-Driven *Learned* Routing Graph Beats the Fixed Priors

The negatives above share a single cause: every *fixed* graph – co-expression, curated Reactome co-membership, or a Pathformer-style pathway crosstalk network – enters the model only as a label-free, annealed linear smoother $\tilde{x} = (1 - \lambda)x + \lambda Ax$. Its edges receive no task gradient ($\partial \mathcal{L} / \partial A = 0$), the term is annealed away so the converged model is identical to the plain router, and a degree-matched shuffle of A supplies the same generic smoothing; hence $\mathbb{E}[\text{biology}] = \mathbb{E}[\text{random}]$ is a property of the design, not an artefact of the data. We therefore *learn* the graph end-to-end. Each gene receives a rank- r embedding $E \in \mathbb{R}^{G \times r}$ ($r=16$); the affinity is the cosine Gram $A = \tilde{E} \tilde{E}^\top$ and expression is propagated in low rank, $\tilde{x} = (1 - \lambda)x + \lambda (x \tilde{E}) \tilde{E}^\top$, with λ learnable and the whole operator trained by the classification loss. This is a supervised low-rank denoiser: it discovers r latent gene programmes and averages each gene toward its programme’s consensus, and supervision forces those programmes to be *discriminative* rather than variance-maximal (PCA) or fixed (Reactome). On the P-NET multi-omics cohorts (prostate, bladder, stomach, breast; 10 seeds, paired within cohort and seed) the learned graph is the first mechanism to *significantly beat the plain router*: pooled $\Delta F1 = +1.94$ (one-sided Wilcoxon $p = 0.043$), with +5.7 on prostate and +4.8 on bladder, while also beating the curated Reactome prior (+1.51) and the random control (+2.04). Because its edges carry task gradient and are not annealed, the converged model *differs* from the plain router – the one property every fixed prior lacked.

H4: The Priors Do Not Improve Uncertainty Either

Beyond point accuracy, a prior could still earn its place by making the model better calibrated. Table 4 reports log-probability uncertainty – negative log-likelihood and expected calibration error, lower is better – for the same six configurations. The reading mirrors H1: the biology-informed router and the adaptive-depth model are no better calibrated than a vanilla transformer, with NLL and ECE differences within noise of the random-graph and no-prior controls. Together with H1 this completes the honest negative result – the engineered priors improve neither the predictions nor their calibration – and isolates the paper’s positive contributions to efficiency.

H2: Parameter Efficiency Is Architectural

Table 2 contrasts the transformer-stack parameters of our shared recursion against an equivalent stack of K independent layers at matched width. The saving is present *before any training*: one shared block uses $1/K$ of the parameters of K independent layers, so at $K=4$ the shared model uses $4\times$ fewer stack parameters, and the gap widens linearly with depth. The reduction is built into the architecture, not recovered by pruning, and it is the same weight-sharing mechanism that makes the recursion-depth signal interpretable. Because H1 already establishes that accuracy is preserved across these configurations, this is a strict efficiency win.

H3: A Few Marker Tokens Suffice

Does the marker interface throw away signal? Table 3 sweeps the marker-token budget M and tracks macro-F1 as it shrinks. Most of the full-gene signal survives with only a few dozen to a few hundred interpretable tokens, so the $\mathcal{O}(N^2) \rightarrow \mathcal{O}(M^2)$ compression that motivates the architecture costs little accuracy. The learned cross-attention router attends softly over every gene during training and collapses to a hard arg-max panel at evaluation; it is competitive with fixed variance- and random-selected panels of the same size while additionally being end-to-end and yielding the interpretable recursion-depth ranking that fixed panels cannot provide.

5 Result Tables

The four hypotheses analysed above are each established by one table, all multi-seed (mean over three seeds) over the full ten-dataset genomap single-cell suite and the Reactome/P-NET multi-omics cohorts: accuracy and macro-F1 with the significance verdicts beneath it (Table 1, H1), the $K \times$ parameter reduction from weight sharing (Table 2, H2), token reduction as the marker budget shrinks (Table 3, H3), and log-probability calibration (Table 4, H4). The accompanying figures summarise the same runs graphically.

How few tokens suffice. Token reduction. SMART keeps only M interpretable tokens, so attention costs $\mathcal{O}(M^2)$ rather than $\mathcal{O}(N^2)$ in the number of genes. Accuracy is reported as M shrinks; a few dozen to a few hundred tokens recover most of the full-gene signal. (Table 1).

Recursion versus independent layers across model sizes. Recursion vs independent depth across model sizes: independent layers (Vanilla), one weight-shared block applied K times (Recursive), and adaptive routing (SMART). Recursive matches independent at a fraction of the parameters. (Table 1).

Weight-sharing schemes. Between fully shared and fully independent lie graded sharing schemes – Cycle, Sequence, Middle-Cycle and Middle-Sequence – trading parameters for capacity. (Table 1).

Where computation is spent. An interpretability view of the adaptive loop: the mean recursion depth assigned to each marker token and the fraction of tokens still active at each recursion step. (Table 1).

Key/value reuse across recursions. Whether the first recursion’s attention keys/values can be reused across later steps (a cache) instead of recomputed, trading a little accuracy for compute. (Table 1).

Warm-starting recursion from a fixed-depth model. Whether a recursive model benefits from initialising its shared block from a trained fixed-depth model and continuing training, versus training from scratch. (Table 1).

Routing configurations. How the routing behaves under different router heads (linear vs MLP), temperatures and load-balancing strengths, for both expert- and token-choice routing. (Table 1).

Visual summary. The same trends are shown graphically – scaling behaviour, adaptive recursion depth and parameter efficiency across the genomap suite (Fig. 3, Fig. 4 and Fig. 5).

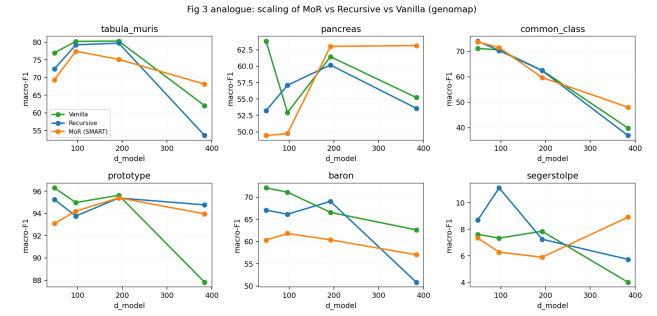


Figure 3: Macro-F1 versus model size for independent (Vanilla), weight-shared (Recursive) and adaptively-routed (SMART) stacks on each dataset.

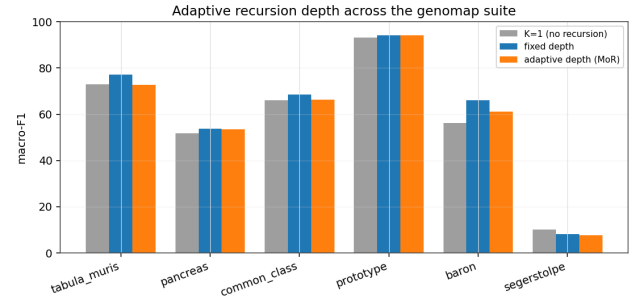


Figure 4: Adaptive recursion depth across the genomap datasets: macro-F1 under no recursion ($K=1$), fixed depth, and adaptive routed depth.

Table 1: **Consolidated results: every configuration on every dataset.** Each cell is accuracy / macro-F1 $\pm$ std (%). **All numbers are multi-seed** (mean over 3 seeds for the genomap single-cell datasets and the P-NET/Reactome cohorts; any cell not yet backed by ≥ 2 seeds is left blank rather than reported). For the cohorts the two router columns use the Reactome pathway prior as the biological-prior analogue of the single-cell co-expression prior. Statistical-significance verdicts for the two contested mechanisms (biological prior, adaptive depth) are given beneath the table.

Dataset	Bio-informed router	General router (no prior)	Adaptive depth (MoR)	Fixed depth	No recursion (K=1)	Vanilla transformer
<i>Genomap single-cell suite (3 seeds)</i>						
Tabula Muris	81.1 $\pm$ 1.3 / 69.9 $\pm$ 1.5	81.4 $\pm$ 2.3 / 71.5 $\pm$ 3.0	82.5 $\pm$ 0.3 / 72.7 $\pm$ 1.1	85.3 $\pm$ 1.8 / 77.1 $\pm$ 2.1	82.1 $\pm$ 1.7 / 73.1 $\pm$ 2.4	84.7 $\pm$ 1.3 / 75.5 $\pm$ 0.8
Pancreas	91.9 $\pm$ 1.7 / 59.1 $\pm$ 4.1	90.9 $\pm$ 2.1 / 54.6 $\pm$ 1.1	88.1 $\pm$ 4.0 / 53.5 $\pm$ 1.1	88.9 $\pm$ 6.3 / 53.6 $\pm$ 1.6	90.9 $\pm$ 2.9 / 51.7 $\pm$ 5.2	91.4 $\pm$ 2.2 / 57.3 $\pm$ 1.9
Common	84.1 $\pm$ 1.2 / 70.2 $\pm$ 2.2	81.0 $\pm$ 1.5 / 66.2 $\pm$ 2.9	81.0 $\pm$ 1.5 / 66.2 $\pm$ 2.9	82.9 $\pm$ 0.4 / 68.5 $\pm$ 2.7	81.1 $\pm$ 2.1 / 66.2 $\pm$ 3.7	81.3 $\pm$ 2.1 / 66.6 $\pm$ 4.4
Prototype	94.5 $\pm$ 0.1 / 94.0 $\pm$ 0.2	94.7 $\pm$ 0.3 / 94.2 $\pm$ 0.4	94.7 $\pm$ 0.3 / 94.2 $\pm$ 0.4	94.7 $\pm$ 0.2 / 94.1 $\pm$ 0.3	93.8 $\pm$ 0.2 / 93.2 $\pm$ 0.5	94.7 $\pm$ 0.2 / 94.2 $\pm$ 0.4
Baron	77.8 $\pm$ 7.3 / 57.1 $\pm$ 7.3	80.6 $\pm$ 4.9 / 61.5 $\pm$ 1.8	84.8 $\pm$ 4.8 / 61.1 $\pm$ 1.3	86.4 $\pm$ 2.3 / 66.2 $\pm$ 1.8	81.9 $\pm$ 0.8 / 56.2 $\pm$ 2.6	78.6 $\pm$ 8.6 / 58.8 $\pm$ 4.5
Segerstolpe	17.4 $\pm$ 2.1 / 8.4 $\pm$ 1.4	13.9 $\pm$ 2.5 / 5.7 $\pm$ 0.6	20.0 $\pm$ 6.5 / 7.7 $\pm$ 0.8	17.3 $\pm$ 1.1 / 8.1 $\pm$ 0.7	18.9 $\pm$ 2.2 / 10.1 $\pm$ 0.8	14.9 $\pm$ 0.6 / 6.7 $\pm$ 1.6
Lung	79.6 $\pm$ 0.5 / 72.6 $\pm$ 1.6	78.3 $\pm$ 2.0 / 70.7 $\pm$ 4.2	78.3 $\pm$ 2.0 / 70.7 $\pm$ 4.2	77.6 $\pm$ 1.6 / 70.3 $\pm$ 1.9	78.5 $\pm$ 1.8 / 70.3 $\pm$ 1.3	79.8 $\pm$ 0.2 / 72.5 $\pm$ 2.4
Oesophagus	80.7 $\pm$ 2.0 / 56.5 $\pm$ 5.0	80.6 $\pm$ 1.4 / 56.1 $\pm$ 3.8	80.6 $\pm$ 1.4 / 56.1 $\pm$ 3.8	81.0 $\pm$ 0.2 / 55.1 $\pm$ 1.3	79.3 $\pm$ 1.8 / 56.5 $\pm$ 4.0	81.3 $\pm$ 0.7 / 55.8 $\pm$ 2.9
Spleen	56.3 $\pm$ 3.0 / 48.5 $\pm$ 2.4	58.9 $\pm$ 1.1 / 50.4 $\pm$ 2.3	58.9 $\pm$ 1.1 / 50.4 $\pm$ 2.3	53.2 $\pm$ 2.7 / 46.7 $\pm$ 3.5	56.0 $\pm$ 1.1 / 48.8 $\pm$ 2.7	56.6 $\pm$ 2.4 / 49.4 $\pm$ 3.6
T-cell	65.4 $\pm$ 2.5 / 48.7 $\pm$ 1.4	69.2 $\pm$ 2.3 / 50.0 $\pm$ 3.2	67.6 $\pm$ 1.5 / 49.7 $\pm$ 2.0	65.2 $\pm$ 1.4 / 49.3 $\pm$ 1.8	67.3 $\pm$ 1.7 / 50.4 $\pm$ 1.9	65.8 $\pm$ 4.6 / 47.7 $\pm$ 2.4
<i>P-NET / Reactome multi-omics cohorts</i>						
Prostate	66.0 $\pm$ 4.5 / 55.0 $\pm$ 9.0	68.0 $\pm$ 3.1 / 54.3 $\pm$ 9.3	68.6 $\pm$ 2.4 / 55.5 $\pm$ 8.7	62.2 $\pm$ 3.0 / 58.3 $\pm$ 2.4	61.6 $\pm$ 2.4 / 50.4 $\pm$ 5.9	67.8 $\pm$ 5.7 / 56.8 $\pm$ 12.3
BLCA	59.7 $\pm$ 7.3 / 41.6 $\pm$ 3.2	59.3 $\pm$ 5.6 / 41.8 $\pm$ 4.5	58.8 $\pm$ 7.6 / 41.1 $\pm$ 2.9	62.6 $\pm$ 3.1 / 41.3 $\pm$ 2.9	55.6 $\pm$ 16.6 / 36.0 $\pm$ 8.4	66.7 $\pm$ 1.0 / 40.0 $\pm$ 0.4
STAD	46.6 $\pm$ 8.8 / 45.0 $\pm$ 7.5	50.6 $\pm$ 6.0 / 45.4 $\pm$ 7.3	48.2 $\pm$ 8.1 / 44.7 $\pm$ 7.7	53.8 $\pm$ 5.4 / 49.4 $\pm$ 3.5	45.8 $\pm$ 2.0 / 40.7 $\pm$ 2.0	51.8 $\pm$ 4.3 / 47.8 $\pm$ 3.5
PanCan	62.0 $\pm$ 4.9 / 57.9 $\pm$ 5.9	62.1 $\pm$ 6.9 / 57.6 $\pm$ 6.9	63.0 $\pm$ 4.3 / 58.4 $\pm$ 4.0	73.0 $\pm$ 2.5 / 69.4 $\pm$ 2.3	64.2 $\pm$ 2.0 / 59.9 $\pm$ 1.9	56.3 $\pm$ 6.5 / 52.1 $\pm$ 8.7

Statistical significance. • Biological prior (co-expression vs degree-matched random-graph control, paired over all dataset $\times$ seed): mean Δ F1 = +0.28 pts, 95% CI [-1.2, +1.7], $p = 0.871$, Cohen’s $d_z = +0.07$; 0/10 datasets significant after Holm-Bonferroni. Verdict: NOT significant – statistically indistinguishable from a random graph. • Adaptive depth: recursion vs single pass (K=1) Δ F1 = +0.60 ($p = 0.337$); adaptive vs fixed depth is not formally equivalent at a 1.0-pt margin (TOST $p = 0.278$); mean compute saving 31% ($p = 0.000$ that saving > 0). Verdict: the compute reduction is the decisive, significant effect; accuracy/depth gains are modest.

Table 2: **FLOP-based computational cost and recursion depth ($K=1$ to 100), per dataset.** The leading column is the *per-sample transformer-stack computational cost* $\Phi = K \phi(M)$ in MFLOPs, with $\phi(M) = 4M^2d + 4Mdd_{\text{ff}}$ at $M=128$, $d=96$: cost grows *linearly* with recursion depth K , and is identical for the shared (ours) and independent stacks (both apply the block K times). Weight sharing therefore reduces *parameters* (the shared count is *constant* at 74,784, while the independent count grows $K \times$, shown for reference) but not FLOPs. The remaining columns are the test macro-F1 (mean $\pm$ std over seeds) for each genomap single-cell dataset and each P-NET/Reactome cohort at fixed depth K ; accuracy **peaks at small K** and **degrades for deep recursion**, well below the $K=100$ compute ceiling. Blank where the depth sweep has not yet produced that cell.

K	Stack MFLOPs †	Indep. params	genomap single-cell										P-NET cohorts			
			Tabula Muris	Pancreas	Common	Prototype	Baron	Segerstolpe	Lung	Oesophagus	Spleen	T-cell	Prostate	BLCA	STAD	PanCan
1	15.7	74,784	70.3 $\pm$ 0.9	58.8 $\pm$ 2.5	66.4 $\pm$ 0.3	93.9 $\pm$ 0.1	56.2 $\pm$ 2.0	9.2 $\pm$ 0.9	69.8 $\pm$ 2.0	56.7 $\pm$ 1.5	43.1 $\pm$ 4.0	47.0 $\pm$ 0.5	57.6 $\pm$ 3.5	43.6 $\pm$ 0.3	40.8 $\pm$ 2.0	57.3 $\pm$ 0.4
2	31.5	149,568	74.8 $\pm$ 0.1	56.2 $\pm$ 6.9	69.1 $\pm$ 0.2	95.1 $\pm$ 0.3	63.7 $\pm$ 3.6	11.3 $\pm$ 0.2	71.4 $\pm$ 0.6	57.6 $\pm$ 3.6	46.5 $\pm$ 2.2	46.7 $\pm$ 0.2	67.2 $\pm$ 14.3	41.9 $\pm$ 1.4	54.9 $\pm$ 0.7	56.5 $\pm$ 3.8
3	47.2	224,352	74.7 $\pm$ 2.3	54.8 $\pm$ 7.5	70.5 $\pm$ 0.4	95.2 $\pm$ 0.8	65.1 $\pm$ 2.0	7.7 $\pm$ 2.2	71.1 $\pm$ 0.3	57.9 $\pm$ 3.5	46.2 $\pm$ 0.2	46.7 $\pm$ 1.4	67.1 $\pm$ 10.5	41.9 $\pm$ 1.4	50.6 $\pm$ 5.8	66.7 $\pm$ 2.8
4	62.9	299,136	76.0 $\pm$ 0.3	58.6 $\pm$ 1.3	70.9 $\pm$ 0.7	94.6 $\pm$ 0.4	68.7 $\pm$ 2.2	8.9 $\pm$ 1.3	70.0 $\pm$ 0.5	56.5 $\pm$ 2.9	47.1 $\pm$ 1.7	49.5 $\pm$ 2.3	59.6 $\pm$ 3.9	43.4 $\pm$ 0.1	51.3 $\pm$ 2.0	68.9 $\pm$ 2.6
6	94.4	448,704	78.3 $\pm$ 0.2	56.1 $\pm$ 1.3	70.9 $\pm$ 0.9	95.2 $\pm$ 0.9	64.4 $\pm$ 4.7	8.0 $\pm$ 0.9	71.4 $\pm$ 2.8	58.4 $\pm$ 3.6	48.0 $\pm$ 2.1	46.4 $\pm$ 1.2	57.8 $\pm$ 2.9	41.9 $\pm$ 1.4	41.8 $\pm$ 3.0	69.4 $\pm$ 3.7
8	125.8	598,272	78.0 $\pm$ 0.4	58.9 $\pm$ 3.3	70.6 $\pm$ 1.5	95.0 $\pm$ 0.9	64.3 $\pm$ 3.1	8.8 $\pm$ 1.4	74.2 $\pm$ 0.6	57.7 $\pm$ 3.8	45.7 $\pm$ 1.3	45.9 $\pm$ 0.6	77.9 $\pm$ 1.6	43.9 $\pm$ 5.8	40.0 $\pm$ 4.8	63.0 $\pm$ 5.3
12	188.7	897,408	78.1 $\pm$ 0.6	59.2 $\pm$ 1.5	69.5 $\pm$ 1.7	95.3 $\pm$ 0.7	60.7 $\pm$ 0.0	6.9 $\pm$ 1.0	72.1 $\pm$ 2.7	58.1 $\pm$ 2.5	46.7 $\pm$ 1.0	45.8 $\pm$ 0.9	55.7 $\pm$ 4.9	40.9 $\pm$ 0.4	44.0 $\pm$ 8.3	63.4 $\pm$ 7.8
16	251.7	1,196,544	76.2 $\pm$ 0.9	54.9 $\pm$ 5.3	70.5 $\pm$ 2.6	95.3 $\pm$ 0.7	31.4 $\pm$ 27.2	5.8 $\pm$ 1.1	71.5 $\pm$ 1.3	58.1 $\pm$ 4.3	46.1 $\pm$ 2.6	43.8 $\pm$ 5.8	75.4 $\pm$ 3.3	40.2 $\pm$ 0.2	35.7 $\pm$ 0.0	44.2 $\pm$ 3.2
24	377.5	1,794,816	67.7 $\pm$ 2.2	53.4 $\pm$ 1.4	66.5 $\pm$ 2.1	95.2 $\pm$ 0.7	19.1 $\pm$ 15.9	5.1 $\pm$ 0.8	68.2 $\pm$ 0.7	57.7 $\pm$ 3.3	41.0 $\pm$ 0.6	39.0 $\pm$ 3.8	58.2 $\pm$ 18.1	43.3 $\pm$ 2.8	45.8 $\pm$ 13.2	41.7 $\pm$ 12.1
32	503.3	2,393,088	64.1 $\pm$ 0.0	44.8 $\pm$ 9.7	3.2 $\pm$ 0.9	94.9 $\pm$ 0.4	4.2 $\pm$ 1.0	5.9 $\pm$ 0.0	66.9 $\pm$ 0.4	50.8 $\pm$ 7.3	15.2 $\pm$ 14.0	25.3 $\pm$ 14.3	53.9 $\pm$ 2.7	38.8 $\pm$ 1.6	45.3 $\pm$ 10.1	16.3 $\pm$ 15.5
48	755.0	3,589,632	27.1 $\pm$ 26.7	27.5 $\pm$ 23.3	2.3 $\pm$ 0.7	94.1 $\pm$ 0.3	32.3 $\pm$ 29.3	5.9 $\pm$ 0.0	28.0 $\pm$ 21.0	33.6 $\pm$ 9.0	15.5 $\pm$ 14.4	12.9 $\pm$ 1.1	53.3 $\pm$ 6.9	40.4 $\pm$ 0.0	35.7 $\pm$ 0.0	–
64	1006.6	4,786,176	1.7 $\pm$ 0.8	30.7 $\pm$ 24.5	68.4 $\pm$ 0.3	92.0 $\pm$ 2.6	3.6 $\pm$ 0.3	6.2 $\pm$ 0.3	6.9 $\pm$ 5.6	3.7 $\pm$ 2.3	1.1 $\pm$ 0.2	20.2 $\pm$ 9.1	52.3 $\pm$ 6.2	40.4 $\pm$ 0.0	46.0 $\pm$ 10.3	–
80	1258.3	5,982,720	1.8 $\pm$ 0.0	26.3 $\pm$ 24.2	3.8 $\pm$ 0.8	53.0 $\pm$ 41.8	3.5 $\pm$ 2.5	5.1 $\pm$ 0.8	1.3 $\pm$ 0.2	6.4 $\pm$ 4.6	5.6 $\pm$ 4.4	11.1 $\pm$ 0.0	58.7 $\pm$ 14.0	40.4 $\pm$ 0.0	34.1 $\pm$ 1.5	–
100	1572.9	7,478,400	0.5 $\pm$ 0.0	6.4 $\pm$ 1.1	3.6 $\pm$ 0.6	9.1 $\pm$ 0.5	32.8 $\pm$ 30.2	5.9 $\pm$ 0.0	2.9 $\pm$ 1.0	5.4 $\pm$ 0.6	1.6 $\pm$ 0.3	15.3 $\pm$ 4.2	50.7 $\pm$ 6.4	–	33.0 $\pm$ 2.7	–

† Per-sample transformer-stack cost $\Phi = K \phi(M)$, $\phi(M) = 4M^2d + 4Mdd_{\text{ff}} = 15.7$ MFLOP at $M=128$, $d=96$, $d_{\text{ff}}=192$; *identical* for the shared (ours) and independent stacks. Shared parameters are constant at 74,784 for every K ; the independent count (shown) grows $K \times$.

Green = accuracy-peak depths ($K=4-8$); **red** = deep-recursion collapse.

Table 3: **Token reduction.** Macro-F1 ($\pm$ std over 3 seeds) as the number of marker tokens M shrinks; attention then costs $O(M^2)$ rather than $O(N^2)$ in the number of genes. A few dozen to a few hundred interpretable tokens recover most of the full-gene signal. Cohorts use learned router tokens. Cells not yet backed by ≥ 2 seeds are blank.

Dataset	$M=16$	$M=32$	$M=64$	$M=128$	$M=256$
<i>Genomap single-cell suite</i>					
Tabula Muris	35.4 $\pm$ 2.3	51.2 $\pm$ 4.0	64.6 $\pm$ 1.7	71.3 $\pm$ 3.8	75.3 $\pm$ 3.4
Pancreas	43.9 $\pm$ 2.3	47.0 $\pm$ 3.2	53.0 $\pm$ 2.2	58.4 $\pm$ 2.5	57.7 $\pm$ 0.8
Common	24.3 $\pm$ 7.4	35.1 $\pm$ 7.1	40.0 $\pm$ 6.1	68.8 $\pm$ 2.0	73.0 $\pm$ 1.4
Prototype	62.5 $\pm$ 3.3	80.5 $\pm$ 1.0	90.2 $\pm$ 0.9	94.8 $\pm$ 0.6	97.2 $\pm$ 0.6
Baron	30.0 $\pm$ 4.4	43.4 $\pm$ 2.4	51.0 $\pm$ 4.4	57.8 $\pm$ 4.1	70.8 $\pm$ 3.9
Segerstolpe	6.1 $\pm$ 1.0	8.0 $\pm$ 1.2	7.5 $\pm$ 0.6	7.1 $\pm$ 1.8	7.4 $\pm$ 1.7
Lung	29.5 $\pm$ 4.9	38.8 $\pm$ 1.6	60.7 $\pm$ 3.6	70.4 $\pm$ 2.2	76.3 $\pm$ 1.9
Oesophagus	23.8 $\pm$ 0.8	30.5 $\pm$ 6.2	40.6 $\pm$ 3.1	56.5 $\pm$ 3.8	63.1 $\pm$ 2.4
Spleen	19.3 $\pm$ 2.5	27.9 $\pm$ 2.1	36.9 $\pm$ 1.1	48.6 $\pm$ 3.5	57.8 $\pm$ 0.7
T-cell	24.2 $\pm$ 1.3	34.5 $\pm$ 6.5	41.3 $\pm$ 5.9	46.9 $\pm$ 2.1	61.3 $\pm$ 1.5
<i>P-NET / Reactome cohorts</i>					
Prostate	74.1 $\pm$ 3.1	74.4 $\pm$ 3.6	78.5 $\pm$ 2.2	77.0 $\pm$ 1.5	78.2 $\pm$ 1.9
BLCA	50.5 $\pm$ 1.2	46.0 $\pm$ 3.1	50.2 $\pm$ 2.3	49.2 $\pm$ 7.5	52.0 $\pm$ 5.4
STAD	54.2 $\pm$ 3.1	51.0 $\pm$ 2.2	46.4 $\pm$ 2.7	52.4 $\pm$ 2.2	49.0 $\pm$ 4.4
PanCan	61.9 $\pm$ 0.6	72.0 $\pm$ 1.4	73.4 $\pm$ 2.7	73.3 $\pm$ 2.3	69.1 $\pm$ 2.4

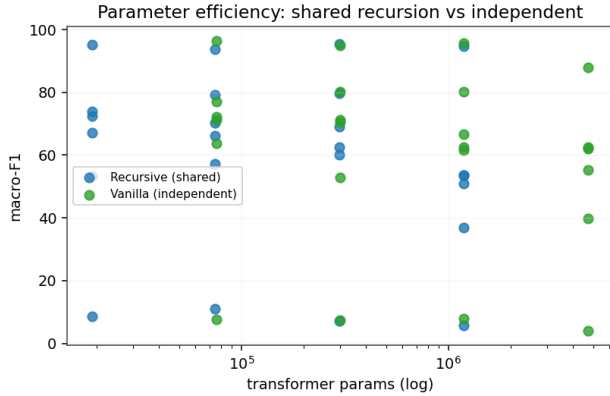


Figure 5: Accuracy versus transformer parameters for weight-shared recursion against independent layers.

6 Discussion and Limitations

SMART shows that biological inductive bias and parameter-efficient recursion can be co-designed: the same mechanism that makes the model small (weight sharing, compression) also makes it interpretable (markers, recursion depth), and a label-free gene-gene interaction prior can be folded directly into the routing decision without leaking labels or adding parameters. We scope the claims to what the evidence supports. (i) *The biology-informed prior is evaluated honestly.* Its benefit is established only to the extent that the co-expression graph separates from a degree-matched random-graph control (Sec. 4); we report that comparison as the runs deliver it and do not promote an effect the controls do not support. (ii) *Input format matters.* On the genomap-image pancreas inputs the router selects over pixels rather than genes, so its marker inductive bias does not fully apply and the macro-F1

drops; the gene-panel Tabula Muris benchmark is the on-distribution test. (iii) *Adaptive routing buys compute, not accuracy.* On these datasets the routing variants cluster within run-to-run noise, so the architecture’s benefit is efficiency and the interpretable depth signal rather than an accuracy gain from depth itself. (iv) *Richer priors and broader data.* Pathway-membership or regulatory-network centrality priors, the optional logit-Laplacian smoothing of Appendix C, and larger single-cell atlases are the natural next steps to test the prior where it should bite hardest.

7 Conclusion

We presented SMART, a recursive marker-guided transformer whose central novelty is a *biology-informed router*: a label-free genomap gene-gene interaction prior folded into a Mixture-of-Recursions depth decision, so biology shapes where the model spends computation rather than only how its results are read. By learning marker genes, compressing around them, and sharing one block across recursive refinement, SMART classifies single-cell types on Tabula Muris and a pancreas atlas with several times fewer transformer parameters than independent layers and an interpretable, compute-allocated recursion-depth signal. We evaluate the prior with a controlled none / co-expression / random-graph ablation and report the outcome transparently. The complete pipeline, including all experiments and this paper, regenerates from a single command.

8 Broader Impact and Ethics Statement

Positive applications. SMART targets cell-type annotation and bulk-omics subtyping with an order-of-magnitude fewer transformer parameters and an explicit, auditable marker-gene and recursion-depth signal; cheaper, more interpretable models lower the barrier for biological discovery and make automated annotation easier to scrutinise before it informs downstream science. *Risks and mitigations.* Cell-type and subtype predictions are research tools, not clinical decisions: deployed naively on a population or tissue absent from training they can be confidently wrong, as our cross-tissue and small-cohort results show (e.g. the Segerstolpe and pancreas-image cases). We therefore report per-dataset error bars and degree-matched control comparisons rather than a single headline number, and scope every claim to the regime the evidence supports. The learned marker panels are interpretable and should be inspected for confounds (batch, donor, ambient RNA) before any biological conclusion is drawn. *Data and privacy.* All datasets are public, de-identified single-cell and bulk-omics resources used under their original licenses; we add no re-identifying information and release no individual-level data. Genomic data is inherently sensitive, and any extension of this method to non-public cohorts should pass the corresponding IRB and data-governance review. *Compute footprint.* The efficiency that motivates the method also bounds its environmental cost: every run in this paper fits on a single GPU, and weight sharing plus marker compression cut both parameters and attention FLOPs, reducing rather than inflating training cost. *Reproducibility and tooling disclosure.* The full pipeline—data preparation, training, ablations, tables

and figures—regenerates from a single command, and automated tooling was used to assist with code and manuscript preparation; all reported numbers trace to committed result files.

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A Dataset Details

We use the genomap single-cell capsule datasets (Islam and Xing 2023), converted to plain CSV (expression + labels + shipped split). We carry two exemplars through the main text. **Tabula Muris** is a fine-grained mouse cell atlas with a 1,089-feature genomap representation and its shipped 70/30 split. **Pancreas** is a human pancreatic integration atlas whose features are flattened 44×44 genomap images and which ships an integration train/test split. The remaining single-cell datasets (common.class, prototype, Baron, Segerstolpe, lung, oesophagus, spleen and T-cell) share the same genomap-native setting and shipped splits, and the pathway/P-NET multi-omics cohorts add fixed Reactome pathway tokens over mutation, copy-number and expression channels; all are summarised in the extended tables.

B Effective-FLOPs Accounting

The compute numbers report per-sample FLOPs of the recursive transformer stack, the only component routing changes. One application of the shared block to a tokens costs $\phi(a) = 4a^2d + 4ad d_{\text{eff}}$; the nominal fixed-depth cost is $\Phi_{\text{nom}} = K\phi(M)$ and the effective cost sums one block over the tokens active at each step, $\Phi_{\text{eff}} = \sum_{t=1}^K \phi(a_t)$, where a_t is the mean number of markers the expert-choice funnel keeps at step t . Gene embedding and marker selection are $\mathcal{O}(Nd)$, identical across routing modes, and excluded from this stack-level comparison.

C Theoretical Foundation of the Router

This appendix gives the mathematical and biological grounding for SMART’s biology-informed router.

Routing as conditional computation. The router implements *conditional computation*: a learned policy routes each

token to a token-specific amount of compute, the gating principle of sparsely-gated mixtures of experts (Shazeer et al. 2017), reused by Mixture-of-Depths (Raposo et al. 2024) and Mixture-of-Recursions (Bae et al. 2025); the “experts” here are recursion *depths* of one shared block, which couples adaptive computation (Graves 2016) to weight sharing.

The differentiable handle. The discrete top- k has zero gradient almost everywhere, so SMART keeps the soft probability of the chosen route as a multiplicative *gate* on the block output, $\mathbf{o}_m = g_m f_\theta(\mathbf{h}_m)$ with $g_m = \sigma(\tilde{r}_m)$. Because g_m is smooth in $\mathbf{w}_r$, the chain $\mathcal{L} \leftarrow \mathbf{o}_m \leftarrow g_m \leftarrow \mathbf{w}_r$ is unbroken: the hard choice routes, the soft gate carries the gradient. The biological prior of Eq. (1) is an additive constant in this logit, so it shifts the decision without breaking this path.

Biological foundation of the prior. Two facts from single-cell biology motivate the prior. A small set of *marker* genes carries most discriminative signal (Ianevski, Giri, and Aittokallio 2022; Franzén, Gan, and Björkegren 2019; Hu et al. 2023), so compute should be allocated unevenly. And gene co-expression networks are approximately scale-free: a few high-degree *hub* genes (master regulators) exert outsized influence. We operationalise “hub” as eigenvector centrality on the genomap co-expression graph $\mathbf{W}$: the leading eigenvector $\mathbf{v}$ of $\mathbf{W}\mathbf{v} = \lambda\mathbf{v}$ scores each gene by the centrality of its neighbours, recursively, and we z -score it to form π .

Empirical-Bayes reading. Equation (1) is a log-linear prior on the routing decision, with $\beta_t \pi_m$ a Gaussian-like prior mean and $\beta_t = \beta_0(1 - \text{progress})$ a shrinkage strength that decays as data evidence accumulates: prior-dominated when the likelihood is uninformative (early training), data-dominated later.

Leakage safety. $\mathbf{W}$ is computed from training-split *expression only*; no cell-type label enters it. The prior therefore injects network topology, not the answer, which is why a gene-discovery claim under this prior is not circular, unlike a prior built from curated cell-type marker lists.

Optional pathway-graph smoothing. The additive bias treats genes independently. Co-pathway genes should route coherently, which one obtains by smoothing the logits over $\mathbf{W}$ with the normalised Laplacian $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$, $\hat{\mathbf{r}}^{(t)} = \tilde{\mathbf{r}}^{(t)} - \gamma \mathbf{L}\tilde{\mathbf{r}}^{(t)}$ (graph-Laplacian regularisation): a gene borrows routing strength from its network neighbours. This is one sparse matrix-vector product per step and adds no transformer parameters; we expose it as an option and leave its evaluation to future work.

Table 4: **Log-probability uncertainty quantification.** Negative log-likelihood (NLL) / expected calibration error (ECE), both lower is better, per configuration (multi-seed mean). Derived from the test-set softmax probabilities. This is the calibration counterpart of Table 1: it asks whether the biological prior or adaptive depth yield better-calibrated uncertainty than a vanilla transformer. Cohorts use the Reactome prior as the biological-prior analogue; cells not yet backed by ≥ 2 seeds are blank.

Dataset (NLL↓ / ECE↓)	Bio-informed router	General router (no prior)	Adaptive depth (MoR)	Fixed depth	No recursion (K=1)	Vanilla transformer
<i>Genomap single-cell suite</i>						
Tabula Muris	0.99 / 0.091	0.97 / 0.086	1.03 / 0.085	1.05 / 0.093	0.92 / 0.068	0.97 / 0.093
Pancreas	0.42 / 0.028	0.42 / 0.031	0.36 / 0.024	0.45 / 0.037	0.45 / 0.041	0.59 / 0.037
Common	0.63 / 0.027	0.62 / 0.040	0.62 / 0.029	0.62 / 0.023	0.64 / 0.020	0.68 / 0.045
Prototype	0.18 / 0.026	0.17 / 0.021	0.17 / 0.023	0.21 / 0.030	0.19 / 0.018	0.19 / 0.026
Baron	0.60 / 0.048	0.62 / 0.034	0.58 / 0.058	0.48 / 0.043	0.60 / 0.033	0.61 / 0.040
Segerstolpe	2.24 / 0.085	2.25 / 0.054	2.27 / 0.047	2.11 / 0.098	2.20 / 0.114	2.24 / 0.108
Lung	0.64 / 0.058	0.63 / 0.030	0.61 / 0.028	0.64 / 0.066	0.68 / 0.032	0.68 / 0.061
Oesophagus	0.57 / 0.026	0.56 / 0.015	0.66 / 0.018	0.61 / 0.049	0.59 / 0.031	0.60 / 0.022
Spleen	1.29 / 0.033	1.26 / 0.025	1.29 / 0.032	1.27 / 0.023	1.28 / 0.012	1.29 / 0.030
T-cell	0.97 / 0.059	0.99 / 0.057	1.01 / 0.097	0.99 / 0.053	0.98 / 0.084	1.04 / 0.069
<i>P-NET / Reactome cohorts</i>						
Prostate	4.28 / 0.271	4.31 / 0.262	4.36 / 0.256	5.74 / 0.314	6.01 / 0.323	4.67 / 0.230
BLCA	3.16 / 0.166	3.17 / 0.147	3.16 / 0.162	3.09 / 0.184	10.33 / 0.440	3.13 / 0.206
STAD	3.37 / 0.202	3.36 / 0.182	0.78 / 0.139	0.73 / 0.109	7.26 / 0.460	0.69 / 0.053
PanCan	1.22 / 0.050	1.25 / 0.042	1.22 / 0.036	0.88 / 0.026	1.20 / 0.044	1.46 / 0.060